

Q No. 1
Question

Two balls A and B of same size are dropped from the same point under gravity. The mass of A is greater than that of B . If the air resistance acting on each ball is same, then

- A) both the balls reach the ground simultaneously
- B) the ball A reaches earlier
- C) the ball B reaches earlier
- D) nothing can be said

Correct Answer

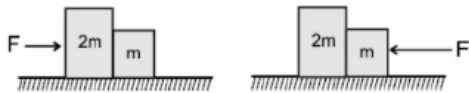
the ball A reaches earlier

Solution

NA

Q No. 2
Question

Two blocks are in contact on a frictionless table. One has mass m and the other $2m$. A force F is applied on $2m$ as shown in the figure. Now the same force F is applied from the right on m . In the two cases respectively, the ratio of force of contact between the two blocks will be :



- A) same
- B) 1: 2
- C) 2: 1
- D) 1: 3

Correct Answer

1: 2

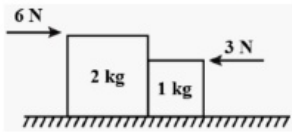
Solution

NA

Q No. 3

Question

Two forces of 6 N and 3 N are acting on the two blocks of 2 kg and 1 kg kept on frictionless floor. What is the force exerted on 2 kg block by 1 kg block?



- A) 1 N
- B) 2 N
- C) 4 N
- D) 5 N

Correct Answer

4 N

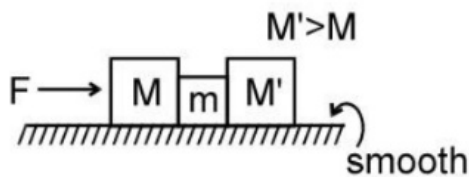
Solution

NA

Q No. 4

Question

A constant force F is applied in horizontal direction as shown. Contact force between M and m is N and between m and M' is N' then



- A) $N = N'$
- B) $N > N'$
- C) $N' > N$
- D) Cannot be determined

Correct Answer

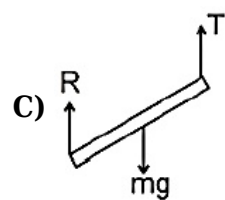
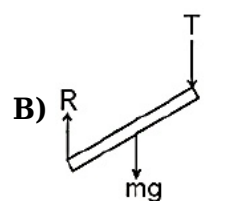
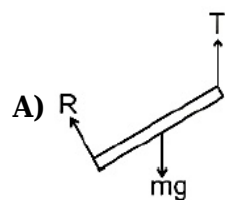
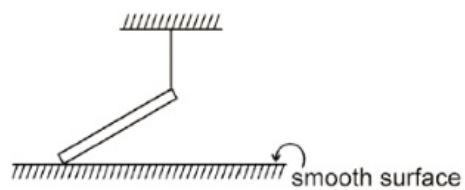
$N > N'$

Solution

NA

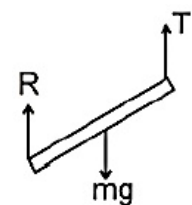
Q No. 5
Question

Which figure represents the correct F.B.D. of rod of mass m as shown in figure :



D) None of these

Correct Answer



Solution

NA

Q No. 6
Question

When a horse pulls a cart, the force needed to move the horse in forward direction is the force exerted by

- A) the cart on the horse
- B) the ground on the horse
- C) the ground on the cart
- D) the horse on the ground

Correct Answer

the ground on the horse

Solution

NA

Q No. 7
Question

Which of the following is an example of a non-contact force?

- A) A child running through a field
- B) A fan blowing a student's hair
- C) A carpenter hammering a nail
- D) A book falling off a desk

Correct Answer

A book falling off a desk

Solution

A non-contact force refers to a force that acts on an object without physically touching it. In the given options, the only example of a non-contact force is "a book falling off a desk." This is because the force of gravity is causing the book to fall without any direct physical contact with another object. The other options involve physical contact between objects, such as a child running through a field, a fan blowing a student's hair, and a carpenter hammering a nail.

Q No. 8
Question

The component of contact force that is perpendicular to the surface in contact is referred to as:

- A) Frictional force
- B) Normal reaction force
- C) Both a and b
- D) None of the mentioned

Correct Answer

Normal reaction force

Solution

NA